

Strength of Materials



Meng2112

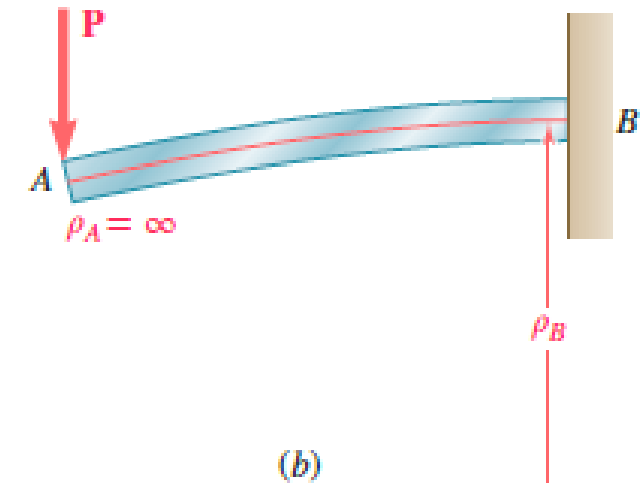
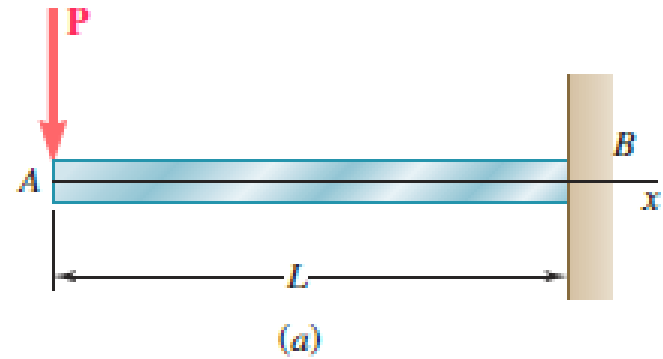
Chapter Five: Deflection of Beams

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Deformation of a Beam Under Transverse Loading



(a) Cantilever beam with concentrated load.

(b) Deformed beam showing curvature at ends.

- Relationship between bending moment and curvature for pure bending remains valid for general transverse loadings.

$$\frac{1}{\rho} = \frac{M(x)}{EI}$$

- Cantilever beam subjected to concentrated load at the free end.

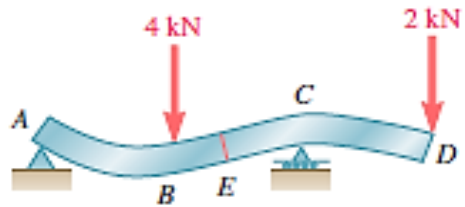
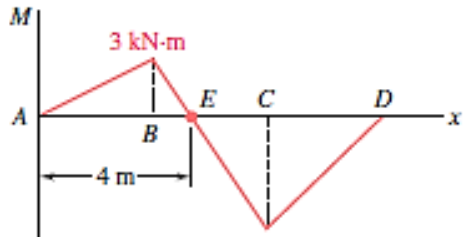
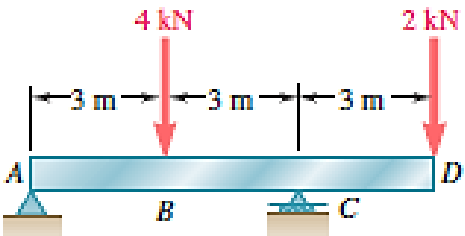
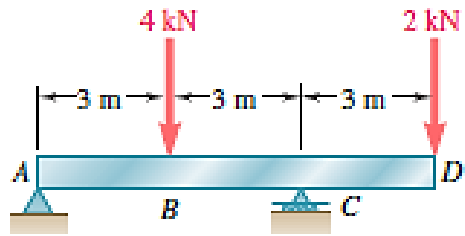
$$\frac{1}{\rho} = -\frac{Px}{EI}$$

- Curvature varies linearly with x

- At the free end A, $\frac{1}{\rho_A} = 0, \quad \rho_A = \infty$

- At the support B, $\frac{1}{\rho_B} \neq 0, \quad |\rho_B| = \frac{EI}{PL}$

Deformation of a Beam Under Transverse Loading



- Overhanging beam
- Reactions at A and C
- Bending moment diagram
- Curvature is zero at points where the bending moment is zero, i.e., at each end and at E.

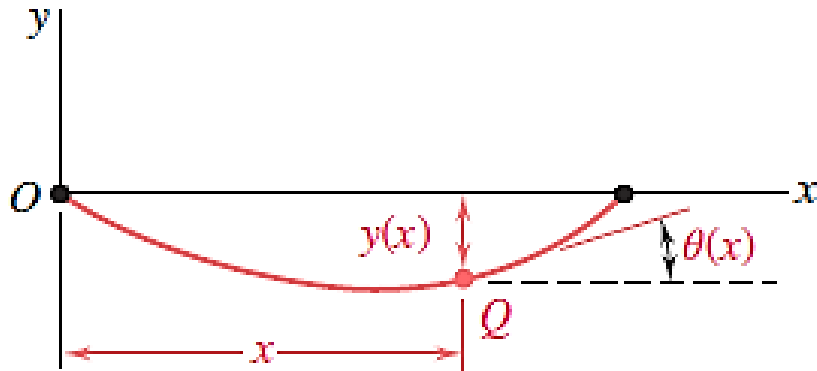
$$\frac{1}{\rho} = \frac{M(x)}{EI}$$

- Beam is concave upwards where the bending moment is positive and concave downwards where it is negative.
- Maximum curvature occurs where the moment magnitude is a maximum.
- An equation for the beam shape or *elastic curve* is required to determine maximum deflection and slope.

deflected shape of the beam, which is represented by its *elastic curve*.

Equation of the Elastic Curve

- From elementary calculus, simplified for beam parameters.



Y= Deflection

Theta(Slope) =dy/dx

$$\frac{1}{\rho} = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}} \quad \frac{1}{\rho} = \frac{d^2y}{dx^2} \rightarrow \frac{d^2y}{dx^2} = \frac{M(x)}{EI}$$

The product EI is called the *flexural rigidity*,

- Substituting and integrating,

$$EI \frac{1}{\rho} = EI \frac{d^2y}{dx^2} = M(x)$$

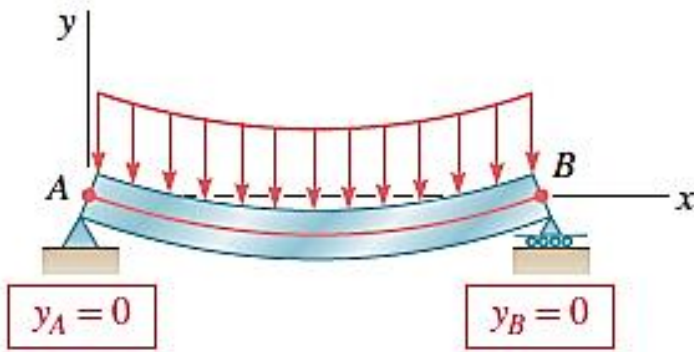
$$EI \theta \approx EI \frac{dy}{dx} = \int_0^x M(x) dx + C_1$$

$$EI y = \int_0^x dx \int_0^x M(x) dx + C_1 x + C_2$$

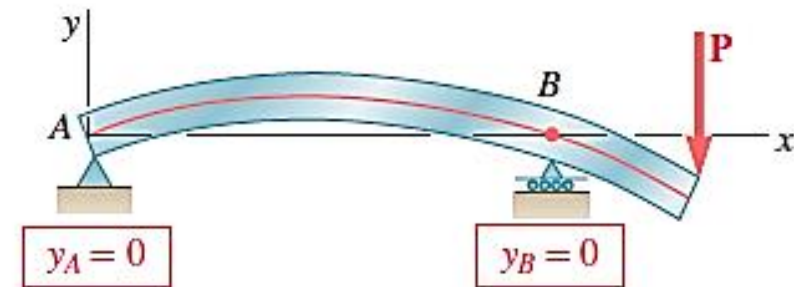
Equation of the Elastic Curve

- Constants are determined from boundary conditions

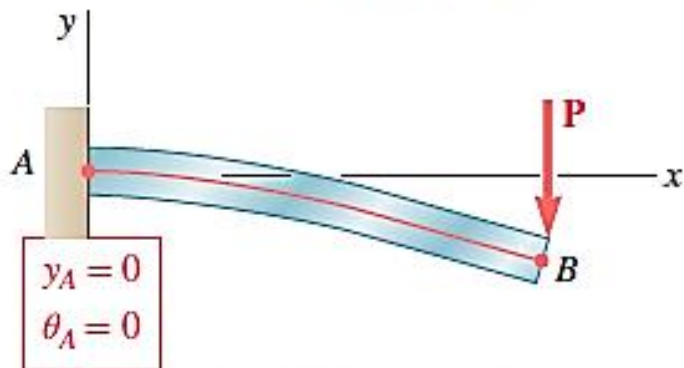
$$EI y = \int_0^x dx \int_0^x M(x) dx + C_1 x + C_2$$



(a) Simply supported beam



(b) Overhanging beam



(c) Cantilever beam

- Three cases for statically determinate beams.

Simply supported beam

$$y_A = 0, \quad y_B = 0$$

Overhanging beam

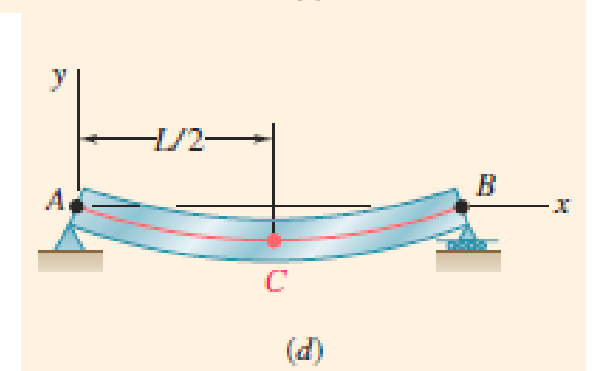
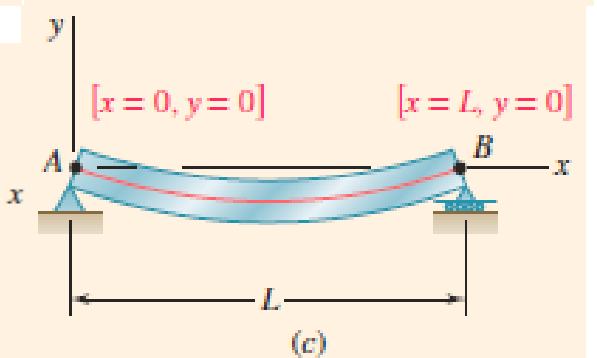
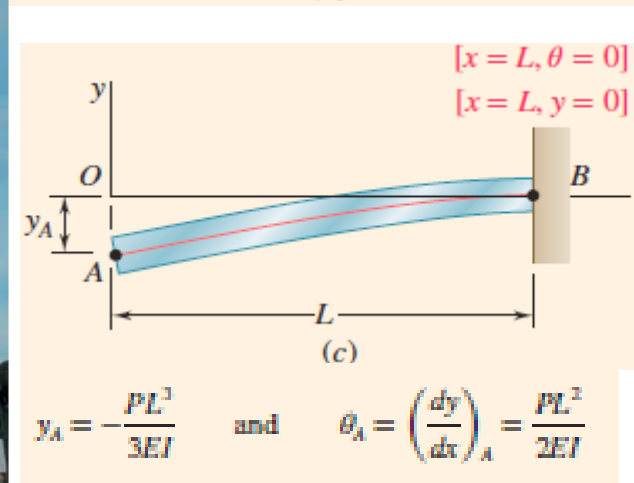
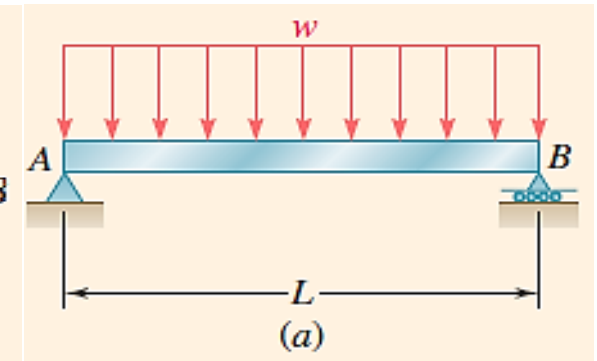
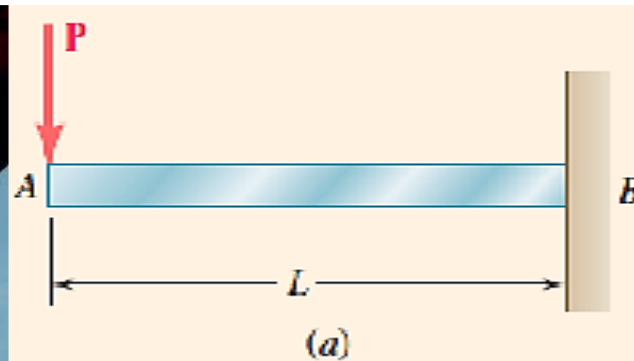
$$y_A = 0, \quad y_B = 0$$

Cantilever beam

$$y_A = 0, \quad \theta_A = 0$$

- More complicated loadings require multiple integrals and application of requirement for continuity of displacement and slope.

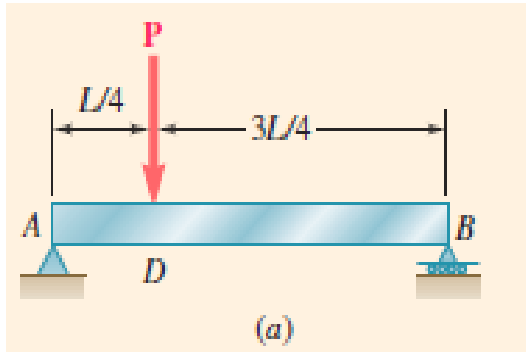
Concept Application (9.1, 9.2, 9.4, 9.5)



$$|y|_{\max} = \frac{5wL^4}{384EI}$$

A different function $M(x)$ would be required in each portion of the beams when a vehicle crosses the completed bridge.

Concept Application 9.3



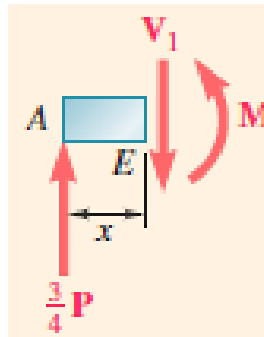
$$R_A = 3/4P$$

$$R_B = P/4$$

(a) Simply supported beam with transverse load P .

- Divide the beam into two portions, AD and DB
- Determine the function $y(x)$ that defines the elastic curve for each of these portions.

1. From A to D ($x < L/4$). Draw the free-body diagram of a portion of beam AE of length $x < L/4$. Take moments about E to obtain



$$M_1 = \frac{3P}{4}x$$

Eq.1

$$EI \frac{d^2y_1}{dx^2} = \frac{3}{4}Px$$

Eq.2

$$\frac{d^2y}{dx^2} = \frac{M(x)}{EI}$$

(b) Free-body diagram of portion AE to find moment left of load P .

where $y_1(x)$ is the function that defines the elastic curve for portion AD of the beam. Integrating in x ,

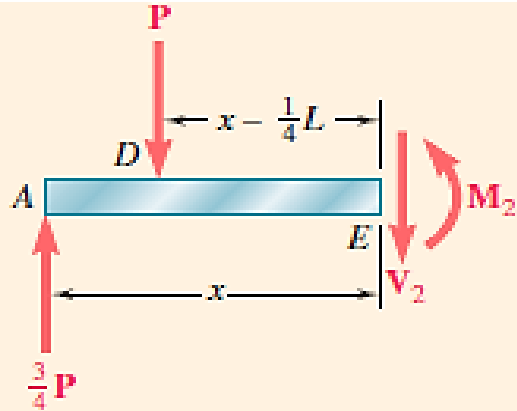
$$EI \theta_1 = EI \frac{dy_1}{dx} = \frac{3}{8}Px^2 + C_1$$

Eq.3

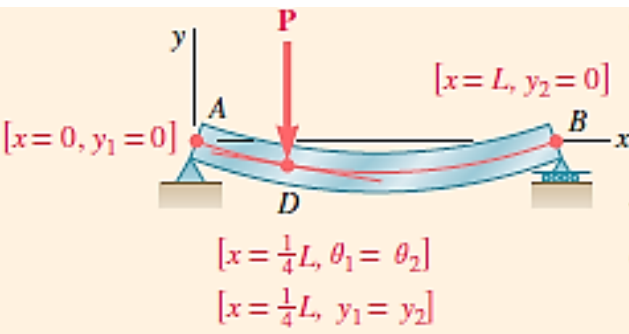
$$EI y_1 = \frac{1}{8}Px^3 + C_1x + C_2$$

Eq.4

Concept Application 9.3



(c) Free-body diagram of portion AE to find moment right of load P .



d) Boundary conditions.

2. From D to B ($x > L/4$). Now draw the free-body diagram of a portion of beam AE of the length $x > L/4$ and write

$$M_2 = \frac{3P}{4}x - P\left(x - \frac{L}{4}\right) \quad \text{Eq.5}$$

rearranging terms

$$EI \frac{d^2 y_2}{dx^2} = -\frac{1}{4}Px + \frac{1}{4}PL \quad \text{Eq.6}$$

$$\frac{d^2 y}{dx^2} = \frac{M(x)}{EI}$$

where $y_2(x)$ is the function that defines the elastic curve for portion DB of the beam. Integrating in x ,

$$EI \theta_2 = EI \frac{dy_2}{dx} = -\frac{1}{8}Px^2 + \frac{1}{4}PLx + C_3 \quad \text{Eq.7}$$

$$EI y_2 = -\frac{1}{24}Px^3 + \frac{1}{8}PLx^2 + C_3x + C_4 \quad \text{Eq.8}$$

Determination of the Constants of Integration.

The conditions satisfied by the constants of integration are summarized.

At the support A, where the deflection is defined by Eq. (4), $x = 0$ and $y_1 = 0$.

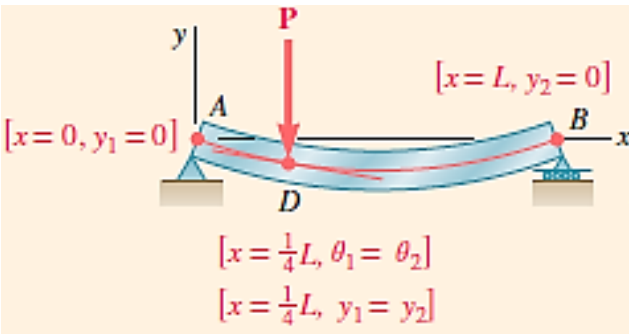
At the support B, where the deflection is defined by Eq. (8), $x = L$ and $y_2 = 0$.

Also, the fact that there can be no sudden change in deflection or in slope at point D requires that $y_1 = y_2$ and $\theta_1 = \theta_2$ when $x = L/4$. Therefore,

Concept Application 9.3

$$EI y_1 = \frac{1}{8}Px^3 + C_1x + C_2 \quad \text{Eq.4}$$

$$EI y_2 = -\frac{1}{24}Px^3 + \frac{1}{8}PLx^2 + C_3x + C_4 \quad \text{Eq.8}$$



Determination of the Constants of Integration.

d) Boundary conditions.

$$EI \theta_1 = EI \frac{dy_1}{dx} = \frac{3}{8}Px^2 + C_1 \quad \text{Eq.3}$$

$$EI \theta_2 = EI \frac{dy_2}{dx} = -\frac{1}{8}Px^2 + \frac{1}{4}PLx + C_3 \quad \text{Eq.7}$$

$$[x = 0, y_1 = 0], \text{Eq. (4):} \quad 0 = C_2$$

Eq.9

$$[x = L, y_2 = 0], \text{Eq. (8):} \quad 0 = \frac{1}{12}PL^3 + C_3L + C_4$$

Eq.10

$$[x = L/4, \theta_1 = \theta_2], \text{Eqs. (3) and (7):}$$

$$\frac{3}{128}PL^2 + C_1 = \frac{7}{128}PL^2 + C_3$$

Eq.11

$$[x = L/4, y_1 = y_2], \text{Eqs. (4) and (8):}$$

$$\frac{PL^3}{512} + C_1\frac{L}{4} = \frac{11PL^3}{1536} + C_3\frac{L}{4} + C_4$$

Eq.12

Solving these equations simultaneously,

$$C_1 = -\frac{7PL^2}{128}, \quad C_2 = 0, \quad C_3 = -\frac{11PL^2}{128}, \quad C_4 = \frac{PL^3}{384}$$

Concept Application 9.3

Substituting for C_1 and C_2 into Eqs. (3) and (4), $x \leq L/4$ is

$$EI \theta_1 = \frac{3}{8}Px^2 - \frac{7PL^2}{128} \quad \text{Eq.13}$$

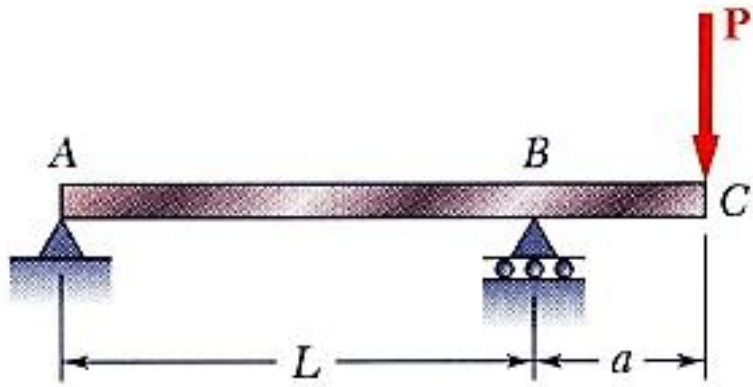
$$EI y_1 = \frac{1}{8}Px^3 - \frac{7PL^2}{128}x \quad \text{Eq.14}$$

Letting $x = L/4$ in each of these equations, the slope and deflection at point D are

$$\theta_D = -\frac{PL^2}{32EI} \quad \text{and} \quad y_D = -\frac{3PL^3}{256EI}$$

Note that since $\theta_D \neq 0$, the deflection at D is *not* the maximum deflection of the beam.

Exercise



$$\begin{array}{lll} W14 \times 68 & I = 723 \text{ in}^4 & E = 29 \times 10^6 \text{ psi} \\ P = 50 \text{ kips} & L = 15 \text{ ft} & a = 4 \text{ ft} \end{array}$$

For portion AB of the overhanging beam

- (a) derive the equation for the elastic curve
- (b) determine the maximum deflection
- (c) evaluate $Y_{\max}$

Solution

- Develop an expression for $M(x)$ and derive differential equation for elastic curve.
- Integrate differential equation twice and apply boundary conditions to obtain elastic curve.
- Locate point of zero slope or point of maximum deflection.
- Evaluate corresponding maximum deflection.

Exercise

Solution

- Develop an expression for $M(x)$ and derive differential equation for elastic curve.

- Reactions:

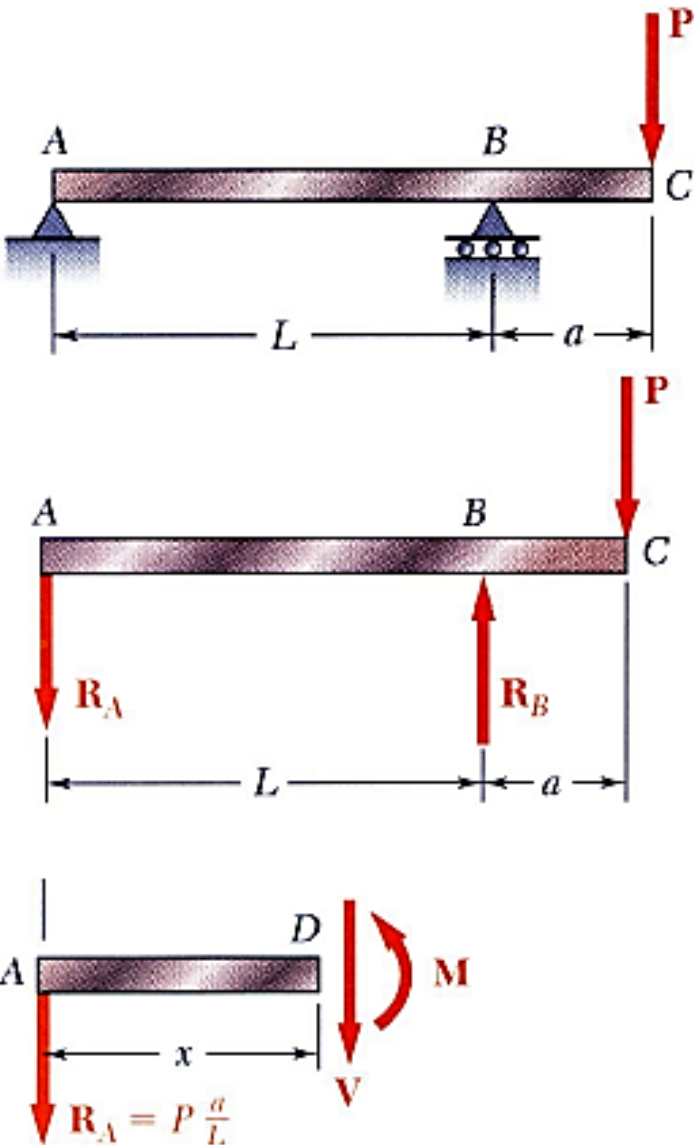
$$R_A = \frac{Pa}{L} \downarrow \quad R_B = P \left(1 + \frac{a}{L} \right) \uparrow$$

- From the free-body diagram for section AD

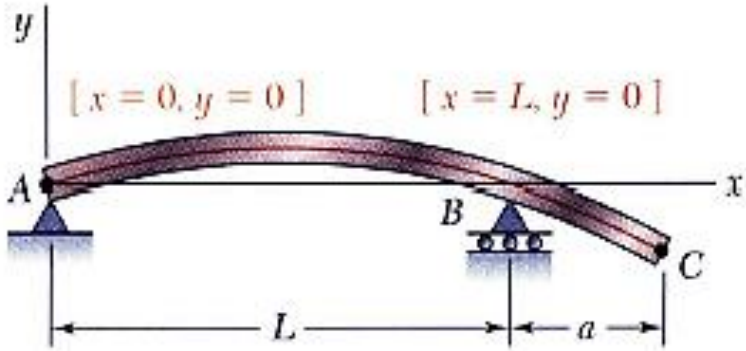
$$M = -P \frac{a}{L} x \quad (0 < x < L)$$

The differential equation for the elastic curve,

$$EI \frac{d^2 y}{dx^2} = -P \frac{a}{L} x$$



Exercise



$$EI \frac{d^2 y}{dx^2} = -P \frac{a}{L} x$$

- Integrate differential equation twice and apply boundary conditions to obtain elastic curve.

$$EI \frac{dy}{dx} = -\frac{1}{2} P \frac{a}{L} x^2 + C_1$$

$$EI y = -\frac{1}{6} P \frac{a}{L} x^3 + C_1 x + C_2$$

$$\text{at } x=0, y=0: C_2 = 0$$

$$\text{at } x=L, y=0: 0 = -\frac{1}{6} P \frac{a}{L} L^3 + C_1 L \quad C_1 = \frac{1}{6} PaL$$

- Substituting

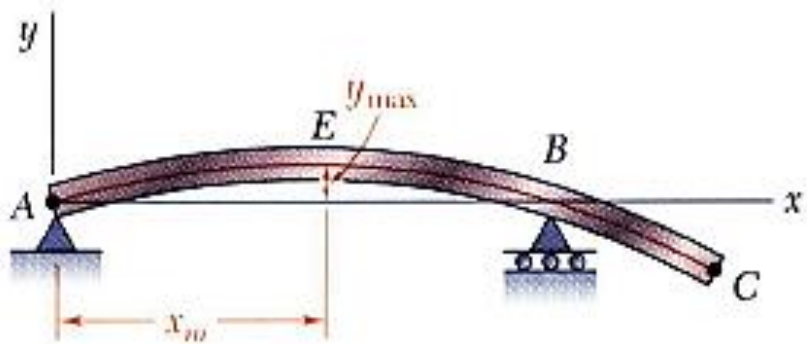
$$EI \frac{dy}{dx} = -\frac{1}{2} P \frac{a}{L} x^2 + \frac{1}{6} PaL \quad \frac{dy}{dx} = \frac{PaL}{6EI} \left[1 - 3 \left(\frac{x}{L} \right)^2 \right]$$

$$EI y = -\frac{1}{6} P \frac{a}{L} x^3 + \frac{1}{6} PaLx$$

$$\boxed{y = \frac{PaL^2}{6EI} \left[\frac{x}{L} - \left(\frac{x}{L} \right)^3 \right]}$$

Equation of Elastic
Curve

Exercise



$$y = \frac{PaL^2}{6EI} \left[\frac{x}{L} - \left(\frac{x}{L} \right)^3 \right]$$

- Locate point of zero slope or point of maximum deflection.

$$\frac{dy}{dx} = 0 = \frac{PaL}{6EI} \left[1 - 3 \left(\frac{x_m}{L} \right)^2 \right] \quad x_m = \frac{L}{\sqrt{3}} = 0.577L$$

- Evaluate corresponding maximum deflection.

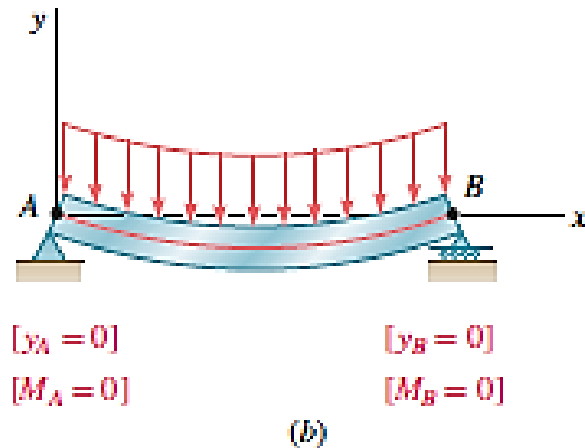
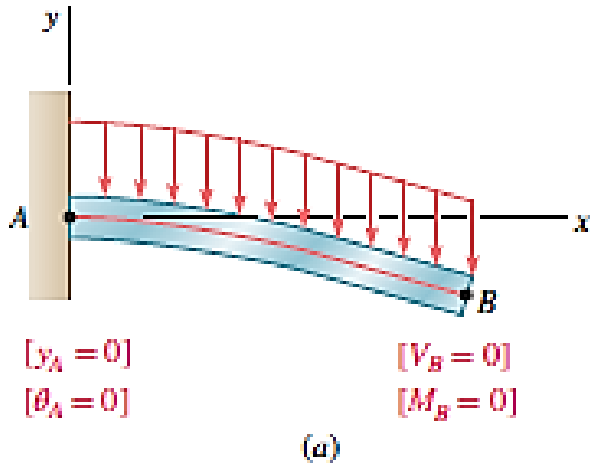
$$y_{\max} = \frac{PaL^2}{6EI} [0.577 - (0.577)^3]$$

$$y_{\max} = 0.0642 \frac{PaL^2}{6EI}$$

$$y_{\max} = 0.0642 \frac{(50 \text{ kips})(48 \text{ in})(180 \text{ in})^2}{6(29 \times 10^6 \text{ psi})(723 \text{ in}^4)}$$

$$y_{\max} = 0.238 \text{ in}$$

Direct Determination of the elastic Curve From the Load Distribution



- For a beam subjected to a distributed load

$$\frac{dM}{dx} = V(x) \quad \frac{d^2M}{dx^2} = \frac{dV}{dx} = -w(x)$$

$$\frac{d^2y}{dx^2} = \frac{M(x)}{EI}$$

- Equation for beam displacement becomes

$$\frac{d^2M}{dx^2} = EI \frac{d^4y}{dx^4} = -w(x)$$

- Integrating four times yields

$$EI \frac{d^4y}{dx^4} = -w(x)$$

$$EI \frac{d^3y}{dx^3} = V(x) = -\int w(x) dx + C_1$$

$$EI \frac{d^2y}{dx^2} = M(x) = -\int dx \int w(x) dx + C_1x + C_2$$

$$EI \frac{dy}{dx} = EI \theta(x) = -\int dx \int dx \int w(x) dx + \frac{1}{2} C_1x^2 + C_2x + C_3$$

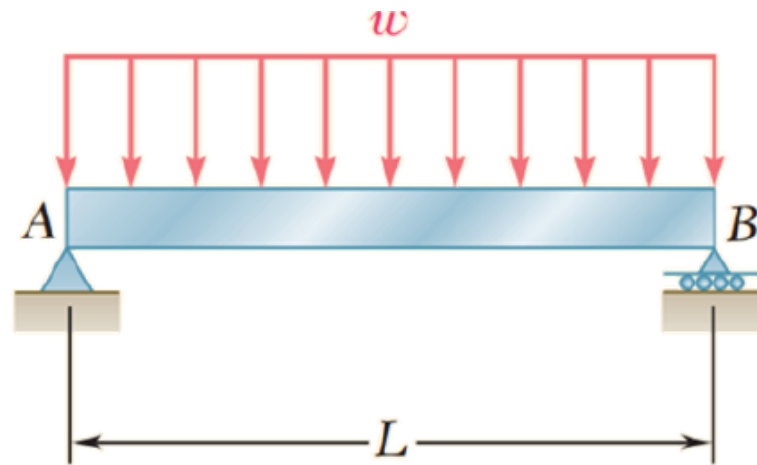
$$EI y(x) = -\int dx \int dx \int dx \int w(x) dx + \frac{1}{6} C_1x^3 + \frac{1}{2} C_2x^2 + C_3x + C_4$$

- Constants are determined from boundary conditions.

Boundary conditions for
(a) cantilever beam
(b) simply supported beam.

Concept Application 9.4

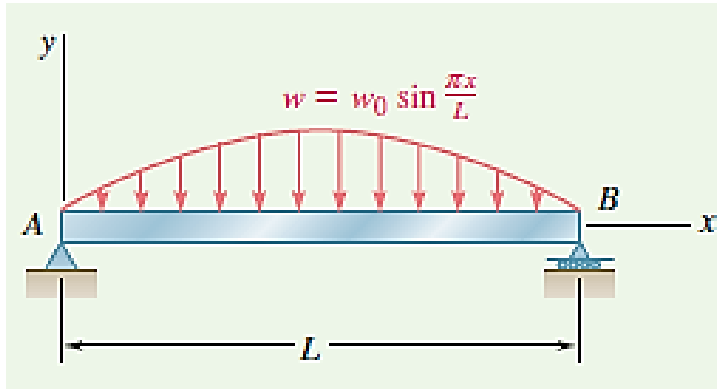
The simply supported prismatic beam AB carries a uniformly distributed load w per unit length. Determine the equation of the elastic curve and the maximum deflection of the beam.



$$\theta = \frac{dy}{dx} = \frac{w}{24EI}(-4x^3 + 6Lx^2 - L^3)$$

$$y = \frac{w}{24EI}(-x^4 + 2Lx^3 - L^3x)$$

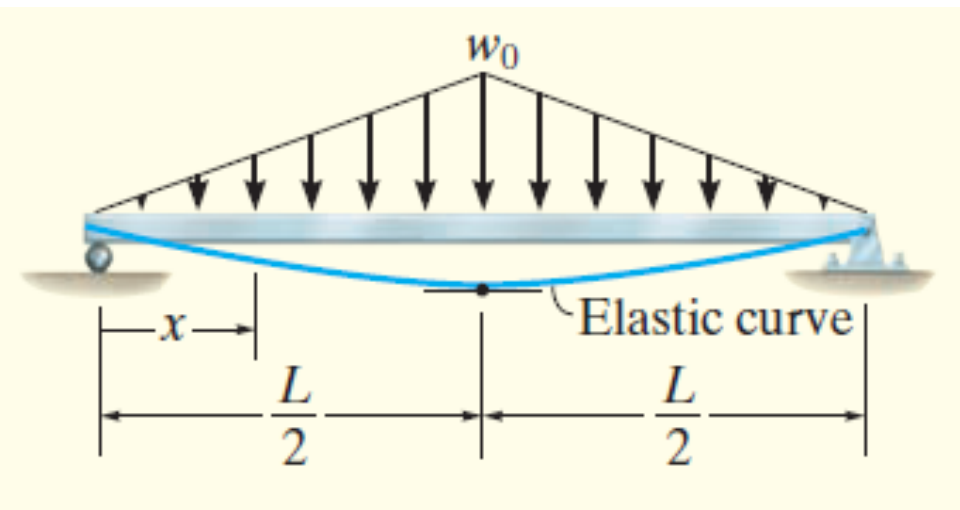
Exercise(HW)



For the beam and loading shown determine

- (a) the equation of the elastic Curve
- (b) the slope at end A
- (c) the maximum deflection.

Review Exercise

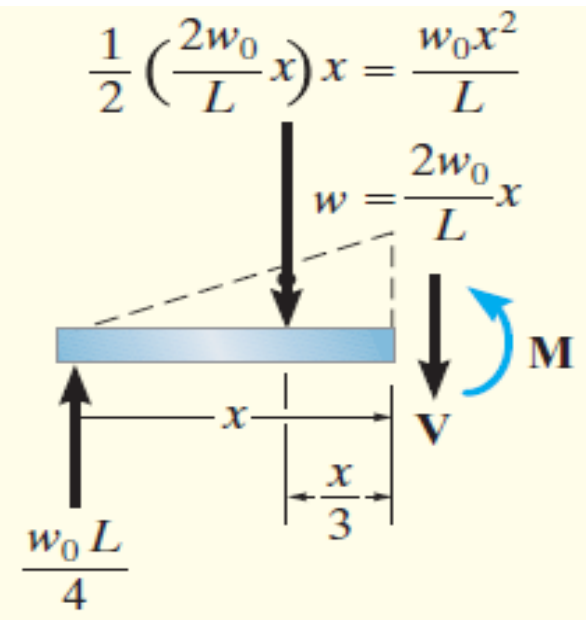


The beam shown in figure supports the triangular distributed loading.

Determine its maximum deflection.
 EI is constant.

SOLUTION

- **Elastic Curve.** Due to symmetry, only one x coordinate is needed for the solution.
In this case $0 < x < L/2$.
- The maximum deflection occurs at the center since the slope is zero at this point.



- **Moment Function.** A free-body diagram of the segment on the left is shown. The equation for the distributed loading is

$$w = \frac{2w_0}{L}x$$

Hence,

$$\begin{aligned} \zeta + \Sigma M_{NA} &= 0; & M + \frac{w_0 x^2}{L} \left(\frac{x}{3} \right) - \frac{w_0 L}{4} (x) &= 0 \\ M &= -\frac{w_0 x^3}{3L} + \frac{w_0 L}{4} x \end{aligned}$$

- **Slope and Elastic Curve.** integrating twice, we have

$$\begin{aligned} EI \frac{d^2 v}{dx^2} &= M = -\frac{w_0}{3L} x^3 + \frac{w_0 L}{4} x \\ EI \frac{dv}{dx} &= -\frac{w_0}{12L} x^4 + \frac{w_0 L}{8} x^2 + C_1 \\ EIV &= -\frac{w_0}{60L} x^5 + \frac{w_0 L}{24} x^3 + C_1 x + C_2 \end{aligned}$$

$$EI \frac{d^2v}{dx^2} = M = -\frac{w_0}{3L}x^3 + \frac{w_0L}{4}x$$

$$EI \frac{dv}{dx} = -\frac{w_0}{12L}x^4 + \frac{w_0L}{8}x^2 + C_1$$

$$EIv = -\frac{w_0}{60L}x^5 + \frac{w_0L}{24}x^3 + C_1x + C_2$$

- The **constants of integration** are obtained by applying the boundary condition $v = 0$ at $x = 0$ and the symmetry condition that $dv/dx = 0$ at $x = L/2$. This leads to

$$C_1 = -\frac{5w_0L^3}{192} \quad C_2 = 0$$

- Hence

$$EI \frac{dv}{dx} = -\frac{w_0}{12L}x^4 + \frac{w_0L}{8}x^2 - \frac{5w_0L^3}{192}$$

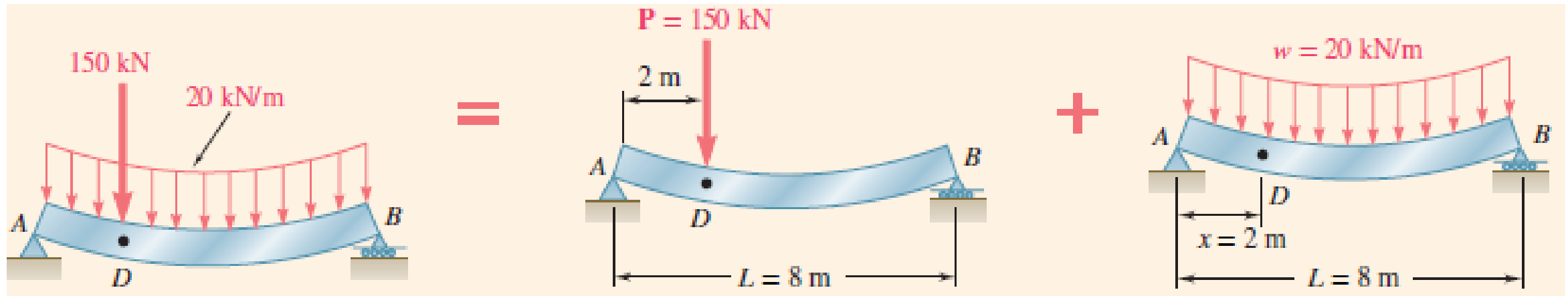
$$EIv = -\frac{w_0}{60L}x^5 + \frac{w_0L}{24}x^3 - \frac{5w_0L^3}{192}x$$

- Determining the maximum deflection at $x = L/2$, we get

$$\theta_A = \frac{5w_0L^3}{192EI} \curvearrowright$$

$$v_{\max} = -\frac{w_0L^4}{120EI}$$

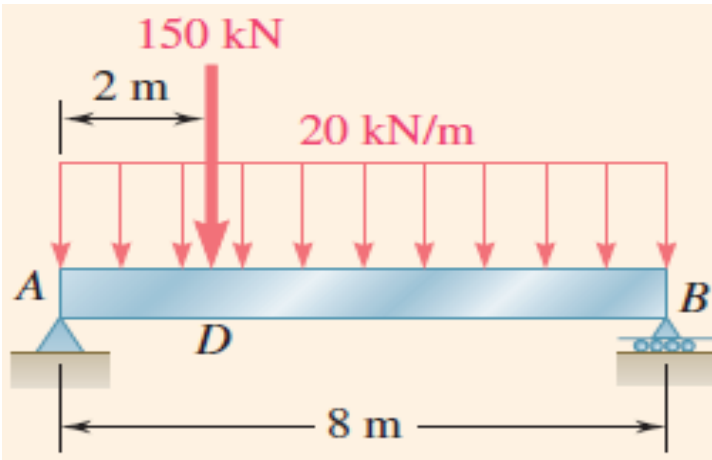
Method of Superposition



Principle of Superposition:

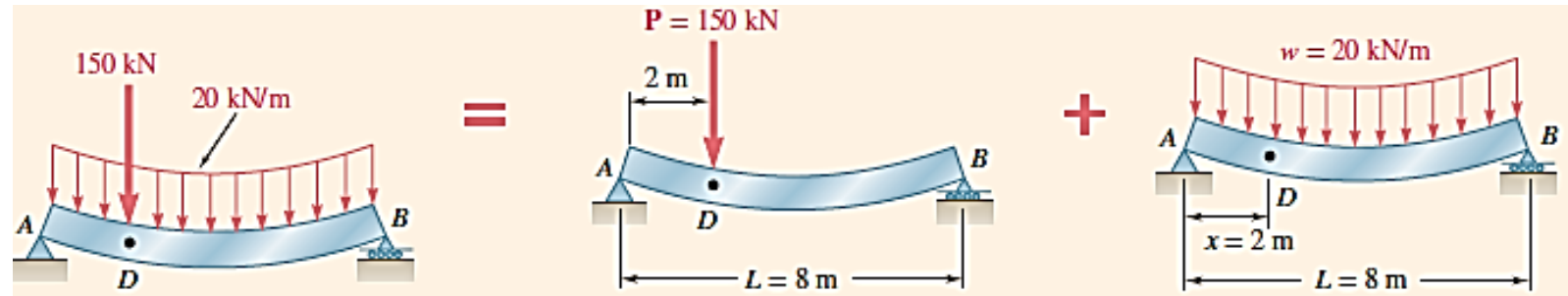
- Deformation of beams subjected to combinations of loadings may be obtained as the linear combination of the deformations from the individual loadings
- Procedure is facilitated by tables of solutions for common types of loadings and supports.

Example



Solution:

The slope and deflection at any point of the beam can be obtained by superposing the slopes and deflections caused by the concentrated load and by the distributed load



Determine the slope and deflection at D for the beam and loading shown, knowing that the flexural rigidity of the beam is $EI = 100 \text{ MN}\cdot\text{m}^2$.

Part I: Concentrated load

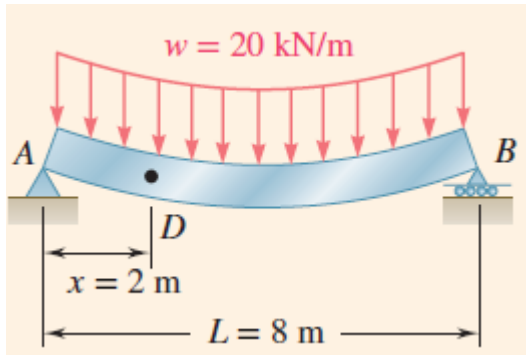
$$(\theta_D)_P = -\frac{PL^2}{32EI} = -\frac{(150 \times 10^3)(8)^2}{32(100 \times 10^6)} = -3 \times 10^{-3} \text{ rad}$$

$$(y_D)_P = -\frac{3PL^3}{256EI} = -\frac{3(150 \times 10^3)(8)^3}{256(100 \times 10^6)} = -9 \times 10^{-3} \text{ m} \\ = -9 \text{ mm}$$

Example

- Part II:

Recalling the equation of the elastic curve obtained for a uniformly distributed load



$$y = \frac{w}{24EI} (-x^4 + 2Lx^3 - L^3x)$$

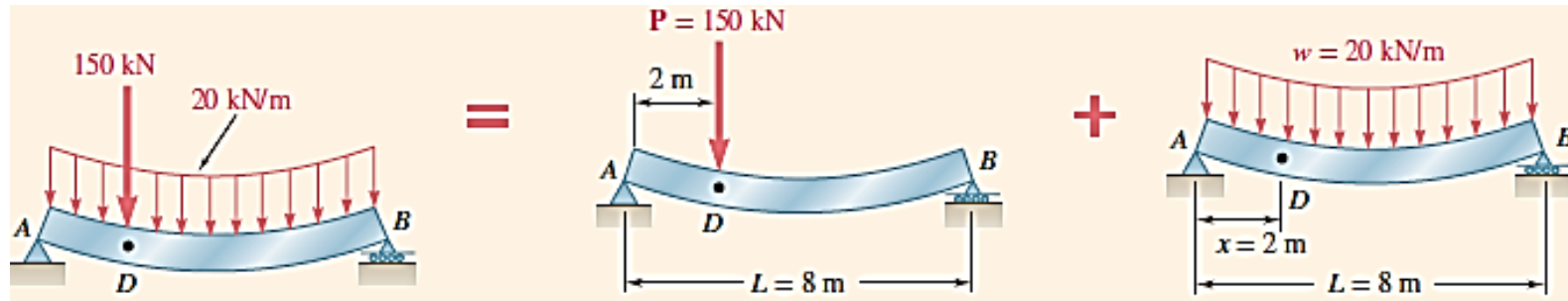
$$\theta = \frac{dy}{dx} = \frac{w}{24EI} (-4x^3 + 6Lx^2 - L^3)$$

- Making $w = 20\text{ kN/m}$, $x = 2\text{ m}$, and $L = 8\text{ m}$

$$(\theta_D)_w = \frac{20 \times 10^3}{24(100 \times 10^6)} (-352) = -2.93 \times 10^{-3} \text{ rad}$$

$$\begin{aligned} (y_D)_w &= \frac{20 \times 10^3}{24(100 \times 10^6)} (-912) = -7.60 \times 10^{-3} \text{ m} \\ &= -7.60 \text{ mm} \end{aligned}$$

Example

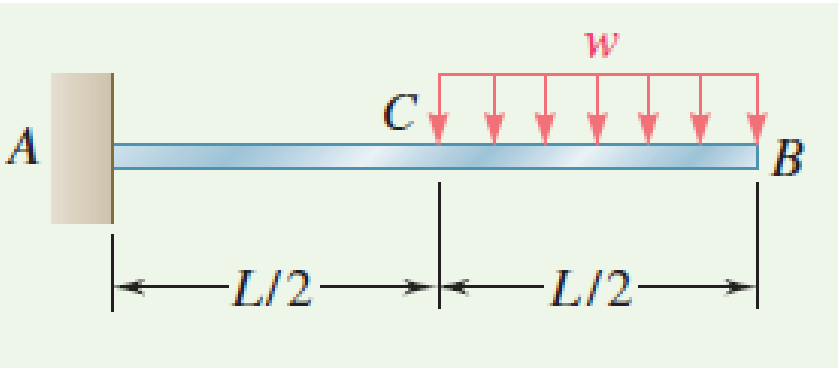


Combining the slopes and deflections produced by the concentrated and the distributed loads,

$$\begin{aligned}\theta_D &= (\theta_D)_P + (\theta_D)_w = -3 \times 10^{-3} - 2.93 \times 10^{-3} \\ &= -5.93 \times 10^{-3} \text{ rad}\end{aligned}$$

$$y_D = (y_D)_P + (y_D)_w = -9 \text{ mm} - 7.60 \text{ mm} = -16.60 \text{ mm}$$

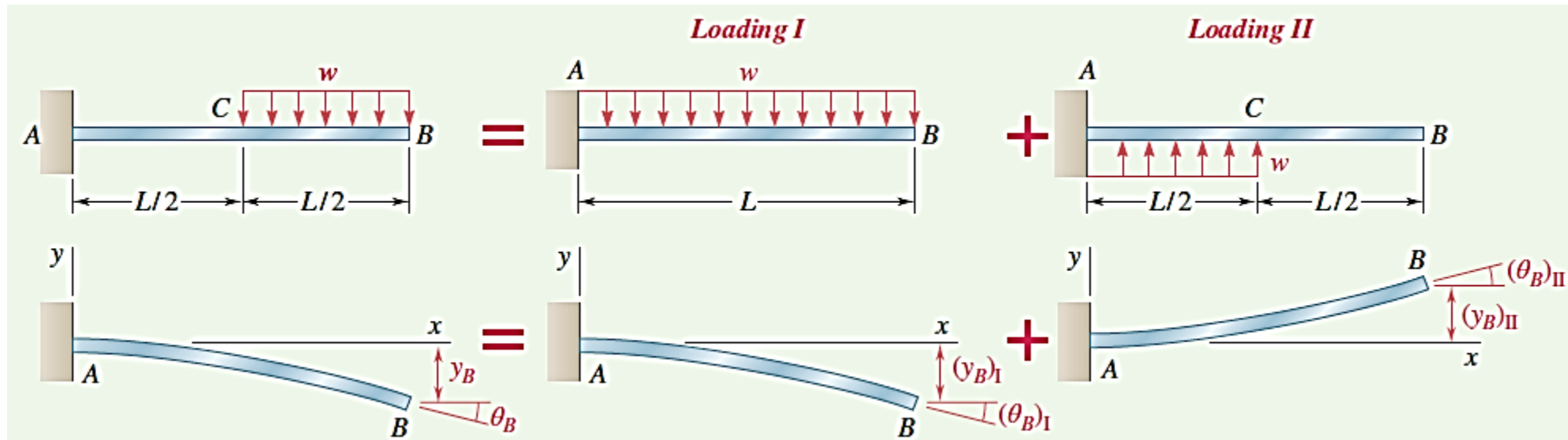
Example



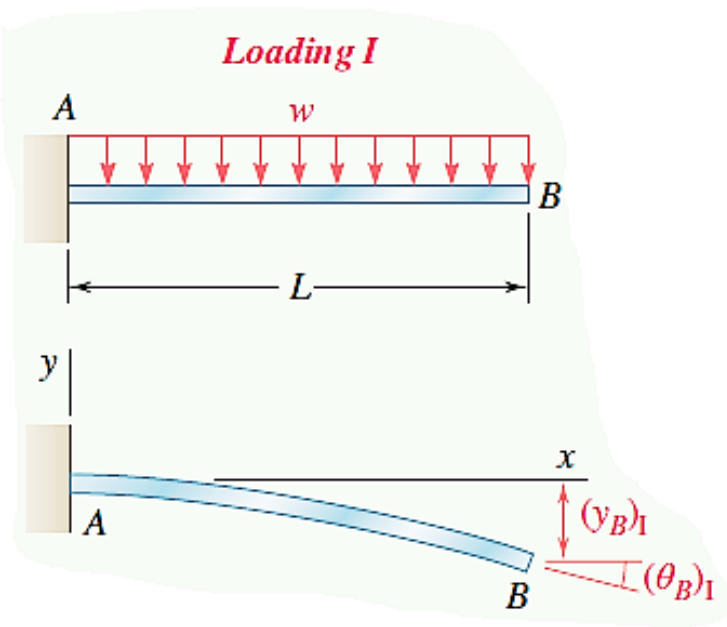
For the beam and loading shown, determine the slope and deflection at point B.

SOLUTION:

Superpose the deformations due to Loading I and Loading II as shown.



Example



Loading I

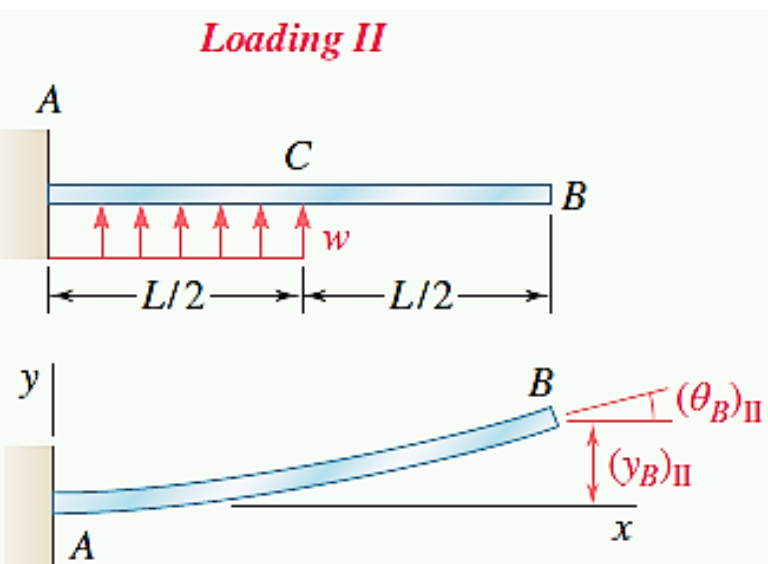
$$(\theta_B)_I = -\frac{wL^3}{6EI}$$

$$(y_B)_I = -\frac{wL^4}{8EI}$$

Loading II

$$(\theta_C)_{II} = \frac{wL^3}{48EI}$$

$$(y_C)_{II} = \frac{wL^4}{128EI}$$

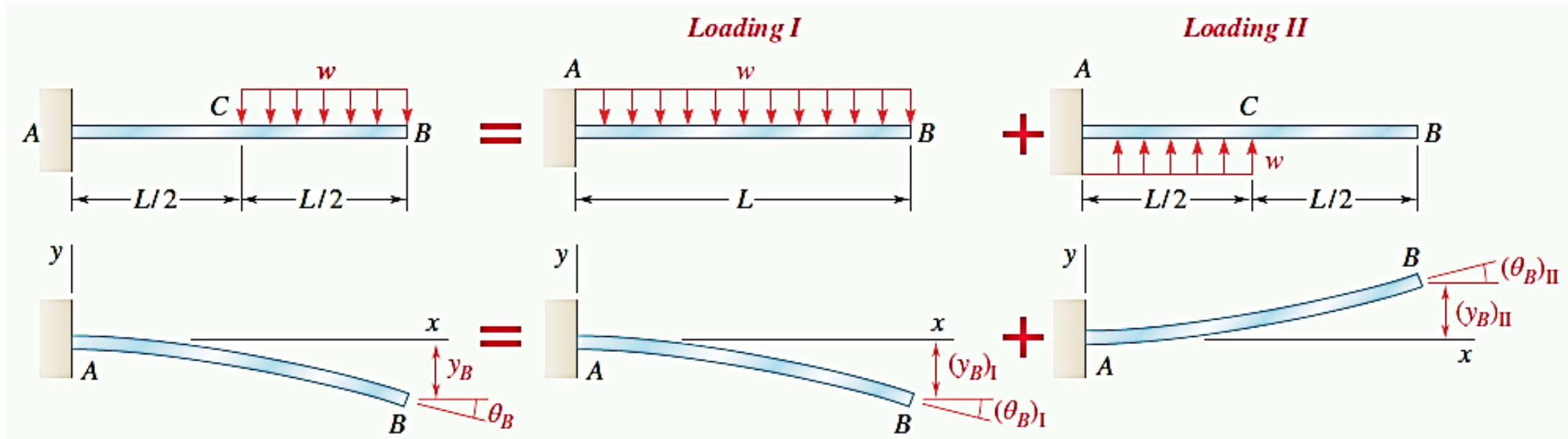


In beam segment CB, the bending moment is zero and the elastic curve is a straight line.

$$(\theta_B)_{II} = (\theta_C)_{II} = \frac{wL^3}{48EI}$$

$$(y_B)_{II} = \frac{wL^4}{128EI} + \frac{wL^3}{48EI} \left(\frac{L}{2} \right) = \frac{7wL^4}{384EI}$$

Example



Combine the two solutions:

$$\theta_B = (\theta_B)_I + (\theta_B)_{II} = -\frac{wL^3}{6EI} + \frac{wL^3}{48EI}$$

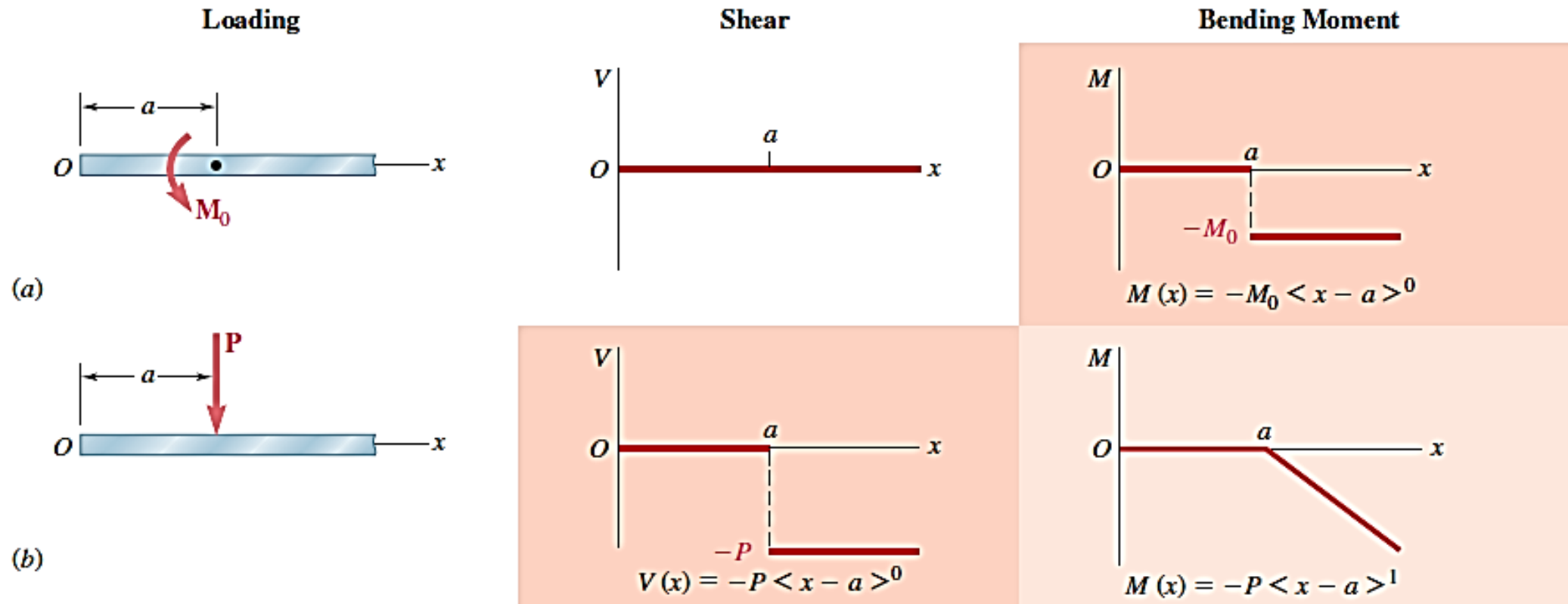
$$\boxed{\theta_B = \frac{7wL^3}{48EI}}$$

$$y_B = (y_B)_I + (y_B)_{II} = -\frac{wL^4}{8EI} + \frac{7wL^4}{384EI}$$

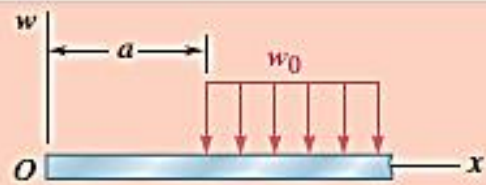
$$\boxed{y_B = \frac{41wL^4}{384EI}}$$

Singularity Functions to Determine the Slope and Deflection

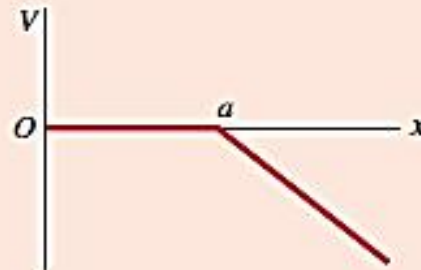
- Singularity functions are used when the bending moment $M(x)$ of a beam is represented by more than one functions.



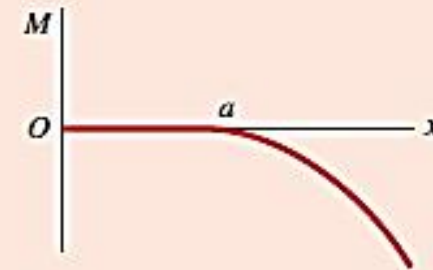
Singularity Functions to Determine the Slope and Deflection



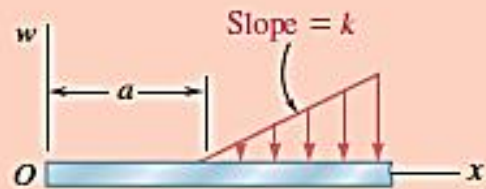
(c) $w(x) = w_0 \langle x - a \rangle^0$



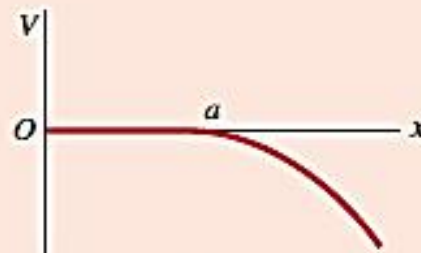
$$V(x) = -w_0 \langle x - a \rangle^1$$



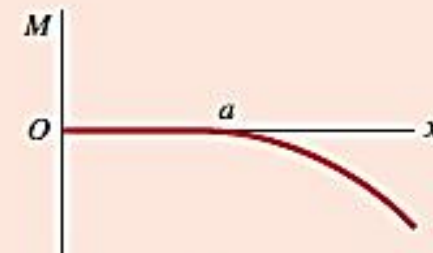
$$M(x) = -\frac{1}{2} w_0 \langle x - a \rangle^2$$



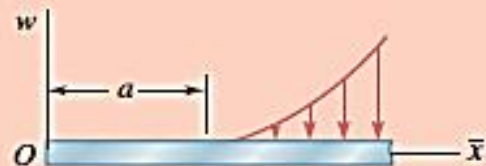
(d) $w(x) = k \langle x - a \rangle^1$



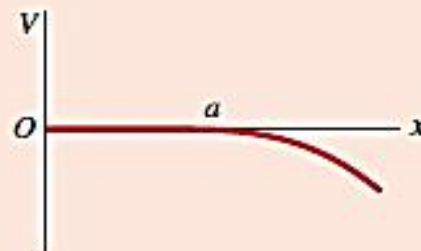
$$V(x) = -\frac{k}{2} \langle x - a \rangle^2$$



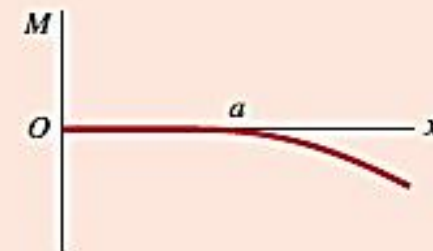
$$M(x) = -\frac{k}{2 \cdot 3} \langle x - a \rangle^3$$



(e) $w(x) = k \langle x - a \rangle^n$



$$V(x) = -\frac{k}{n+1} \langle x - a \rangle^{n+1}$$



$$M(x) = -\frac{k}{(n+1)(n+2)} \langle x - a \rangle^{n+2}$$

Consider the beam and loading shown in Fig. and draw the free-body diagram.

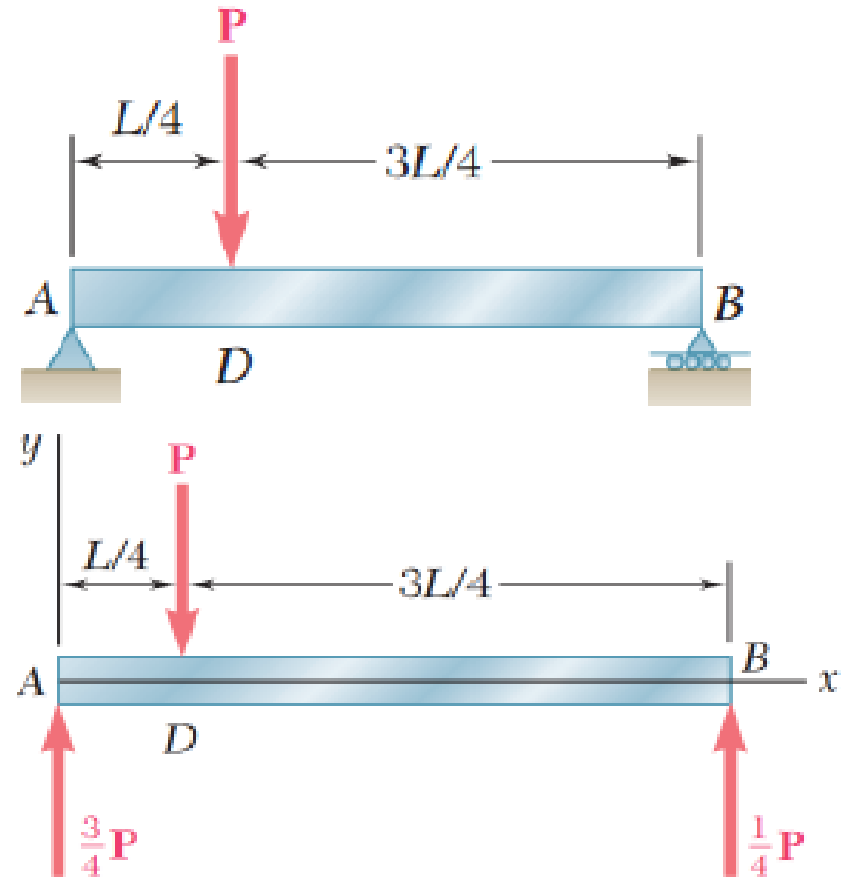
- The singularity functions to represent the contribution to the shear and moment of the concentrated load P can be written as;

$$V(x) = \frac{3P}{4} - P\langle x - \frac{1}{4}L \rangle^0 \quad M(x) = \frac{3P}{4}x - P\langle x - \frac{1}{4}L \rangle$$

$$\text{Hence } EI \frac{d^2 y}{dx^2} = \frac{3P}{4}x - P\langle x - \frac{1}{4}L \rangle$$

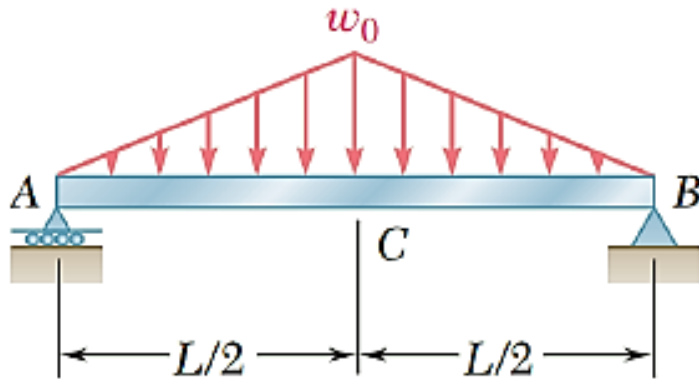
$$EI\theta = EI \frac{dy}{dx} = \frac{3}{8}Px^2 - \frac{1}{2}P\langle x - \frac{1}{4}L \rangle^2 + C_1$$

$$EIy = \frac{1}{8}Px^3 - \frac{1}{6}P\langle x - \frac{1}{4}L \rangle^3 + C_1x + C_2$$



C_1 and C_2 can be found from boundary conditions. Any bracket containing a negative quantity is equal to zero.

Example

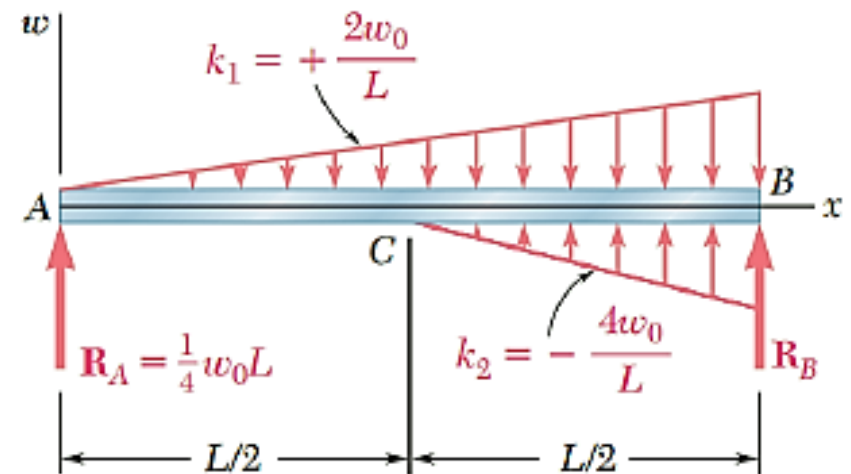


For the prismatic beam and loading shown, determine (a) the equation of the elastic curve, (b) the slope at A, (c) the maximum deflection.

SOLUTION:

Reactions. The total load is $\frac{1}{2}w_0L$. Due to symmetry, each reaction is equal to half that value as $\frac{1}{4}w_0L$.

The given distributed loading is replaced by two equivalent open-ended loadings as shown



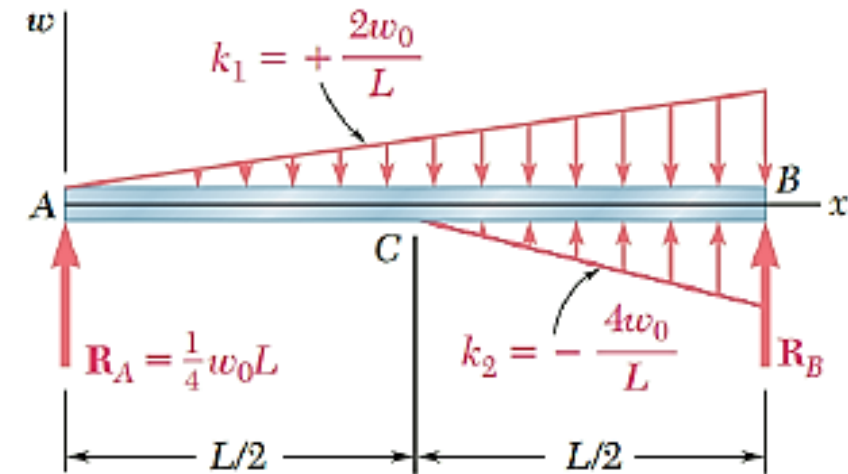
Example

$$w(x) = k_1 x + k_2 \langle x - \tfrac{1}{2}L \rangle = \frac{2w_0}{L}x - \frac{4w_0}{L} \langle x - \tfrac{1}{2}L \rangle$$

Equations for Shear and Bending Moment.

$$V(x) = -\frac{w_0}{L}x^2 + \frac{2w_0}{L} \langle x - \tfrac{1}{2}L \rangle^2 + \tfrac{1}{4}w_0L$$

$$M(x) = -\frac{w_0}{3L}x^3 + \frac{2w_0}{3L} \langle x - \tfrac{1}{2}L \rangle^3 + \tfrac{1}{4}w_0Lx$$



$$EI \frac{d^2 y}{dx^2} = -\frac{w_0}{3L}x^3 + \frac{2w_0}{3L} \langle x - \tfrac{1}{2}L \rangle^3 + \tfrac{1}{4}w_0Lx \quad (1)$$

and, integrating twice in x ,

$$EI \theta = -\frac{w_0}{12L}x^4 + \frac{w_0}{6L} \langle x - \tfrac{1}{2}L \rangle^4 + \frac{w_0L}{8}x^2 + C_1 \quad (2)$$

$$EI y = -\frac{w_0}{60L}x^5 + \frac{w_0}{30L} \langle x - \tfrac{1}{2}L \rangle^5 + \frac{w_0L}{24}x^3 + C_1x + C_2 \quad (3)$$

Equation of the Elastic Curve.

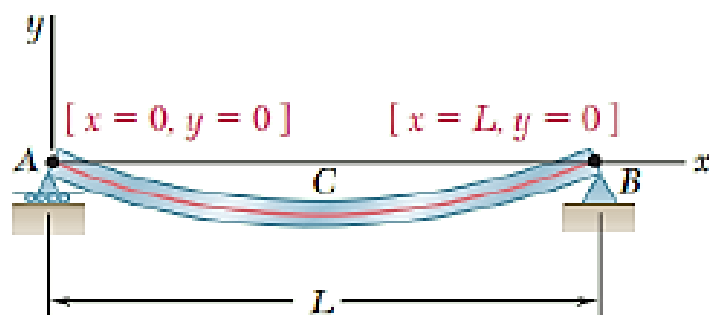


Fig. 2 Boundary conditions.

Boundary Conditions. Referring to Fig. 2,

$[x = 0, y = 0]$: Using Eq. (3) and noting that each bracket $\langle \quad \rangle$ contains a negative quantity and is equal to zero, $C_2 = 0$.

$[x = L, y = 0]$: Again using Eq. (3),

$$0 = -\frac{w_0 L^4}{60} + \frac{w_0}{30L} \left(\frac{L}{2}\right)^5 + \frac{w_0 L^4}{24} + C_1 L \quad C_1 = -\frac{5}{192} w_0 L^3$$

Substituting C_1 and C_2 into Eqs. (2) and (3),

$$EI \theta = -\frac{w_0}{12L} x^4 + \frac{w_0}{6L} \langle x - \tfrac{1}{2}L \rangle^4 + \frac{w_0 L}{8} x^2 - \frac{5}{192} w_0 L^3 \quad (4)$$

$$EI y = -\frac{w_0}{60L} x^5 + \frac{w_0}{30L} \langle x - \tfrac{1}{2}L \rangle^5 + \frac{w_0 L}{24} x^3 - \frac{5}{192} w_0 L^3 x \quad (5) \quad \blacktriangleleft$$

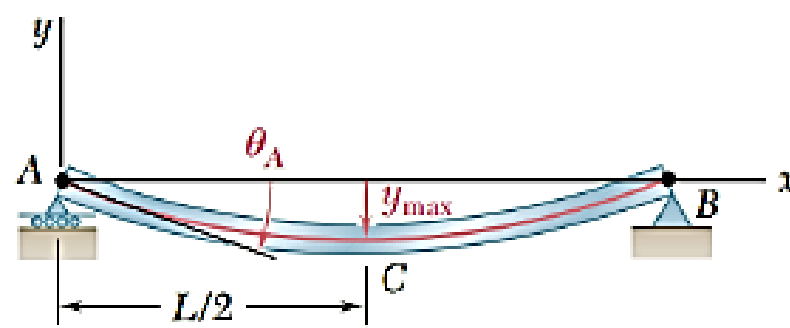


Fig. 3 Deformed elastic curve showing slope at A and maximum deflection at C.

b. Slope at A (Fig. 3). Substituting $x = 0$ into Eq. (4),

$$EI \theta_A = -\frac{5}{192} w_0 L^3 \quad \theta_A = \frac{5w_0 L^3}{192EI} \curvearrowright \blacktriangleleft$$

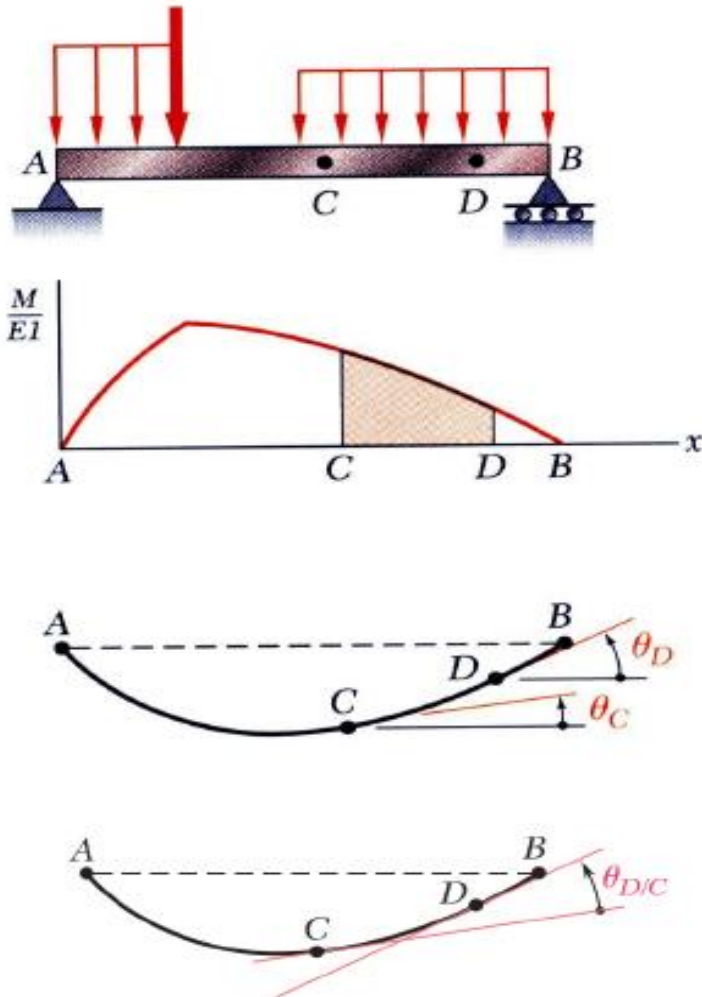
c. Maximum Deflection (Fig. 3). Because of the symmetry of the supports and loading, the maximum deflection occurs at point C where $x = \frac{1}{2}L$. Substituting into Eq. (5),

$$EI y_{\max} = w_0 L^4 \left[-\frac{1}{60(32)} + 0 + \frac{1}{24(8)} - \frac{5}{192(2)} \right] = -\frac{w_0 L^4}{120}$$

$$y_{\max} = \frac{w_0 L^4}{120EI} \downarrow \blacktriangleleft$$

Moment-Area Method

Another Method: Reading



- Geometric properties of the elastic curve can be used to determine deflection and slope.
- Consider a beam subjected to arbitrary loading,

$$\frac{d\theta}{dx} = \frac{d^2y}{dx^2} = \frac{M}{EI}$$

$$\int_{\theta_C}^{\theta_D} d\theta = \int_{x_C}^{x_D} \frac{M}{EI} dx$$

$$\theta_D - \theta_C = \int_{x_C}^{x_D} \frac{M}{EI} dx$$

- *First Moment-Area Theorem:*

$$\theta_{D/C} = \text{area under } (M/EI) \text{ diagram between } C \text{ and } D.$$



Thank you

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